Performance Statistics

Rebuilding Time

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28 November, 2023

When conducting Management Strategy Evaluation objective need to be agreed and then quantified in the form of statistics known as **Performance Statistics**.

Ensuring sustainability requires preventing the stock from becoming overfished, so there is a low probability of compromising productivity. Therefore, many fishery management bodies define a biomass limt reference (B_{lim}) above which the stock should be maintained with high probability. Biomass limits should be based exclusively on the biology of the stock and its resilience to fishing pressure. They should not consider economic factors, as the objective is to reduce risks from a biological perspective, and must integrate dynamic processes such as growth, fecundity, recruitment, mortality and connectivity into indices for exploitation level and spawning reproductive potential.

A way of doing this consistently across stocks and model assumptions is to calculate the rebuilding relative to generation time for a depleted stock to a target level, e.g. the biomass that will support the maximum sustainable yield (B_{MSY}). For example as required for depleted fisheries in the United States. This is also consistent with the IUCN Red List of Threatened Species (Hilton-Taylor and Brackett 2000), which uses generation time as a measure of population decline, as it allows the same criteria to be used for species with extremely different life spans (Mace et al. 2008).

Therefore, as a performance statistic for safety we propose the biomass from which a stock can recover to B_{MSY} within a given time scale and level of fishing mortality ($B_{rebuild}$). The details of which determine the level of risk and so are a management decision. For example, if the management objective is that a depleted stock should recover within a generation time with zero fishing, the corresponding depletion level provides the limit reference point.

In an MSE the 5% level could be used for evaluation, i.e. a strategy is only successful if it keeps a stock above $B_{rebuild}$ with a 95% probability.

Example using 'FLBRP'

For a stock the 'FLBRP' object can be derived, this holds the expected values for the mortality, selectivity, mass, marurity-at-age, and the stock recruitment relationship, and can be used to calculate the expected, i.e. equilibrium values of biomass for any given level of fishing mortality.

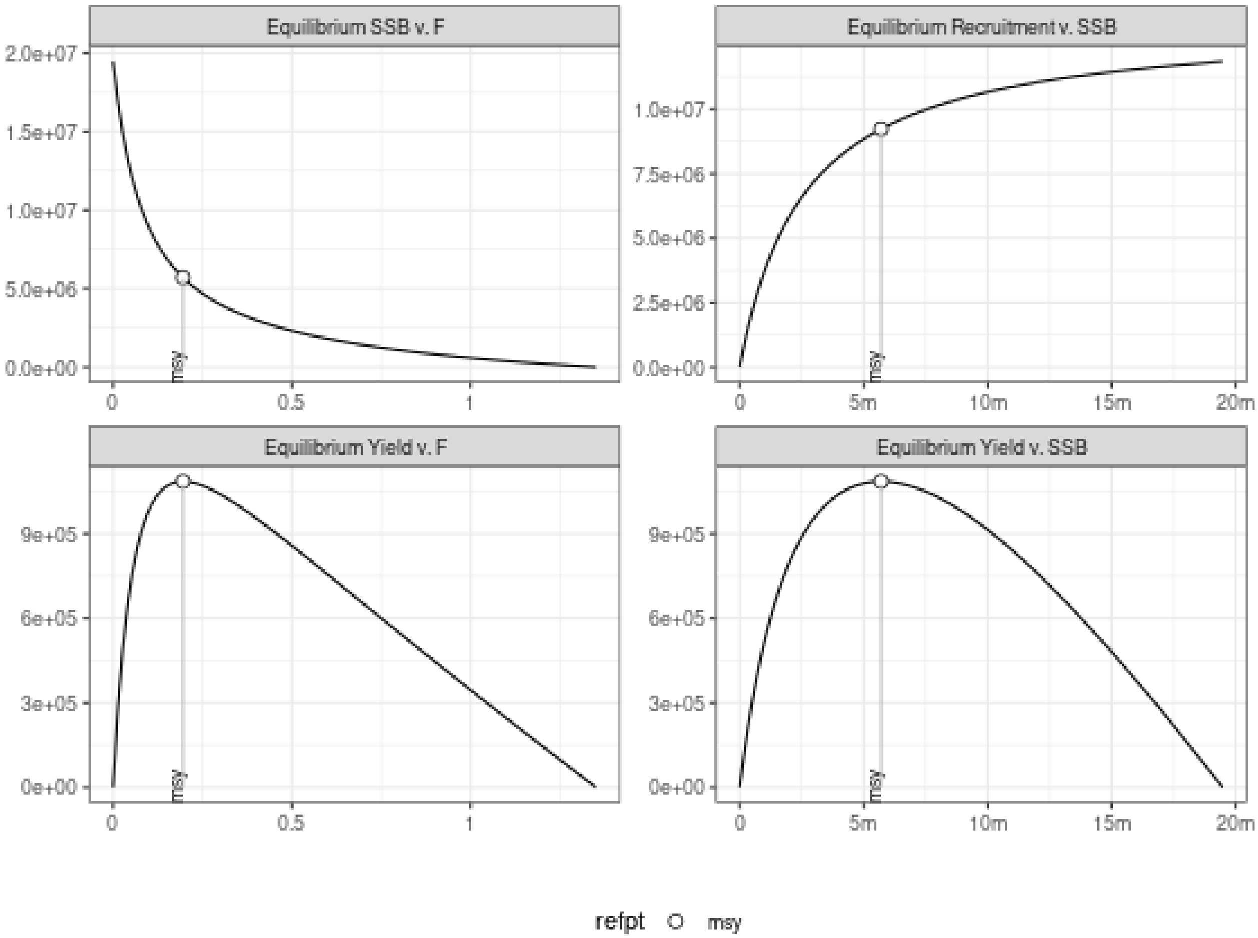


Figure 1 Equilibrium curves showing estimates of SSB, recruitment yield and F.

The 'FLBRP' object is then used to simulate 'FLStock' objects that can be projected through time for a given level of fishing mortality. In **Figure 2** the stock is initially at different levels of depletion and then projected for $F = 0$. The stock then recovers, the dashed horizontal line is SSB/B_{MSY}, and the green vertical line is half a geneartion time, and the red vertical line one generation time. For a desired time for recovery the level that the stock should not fall below can be read from the curves, and is summarised in **Figure 3**.

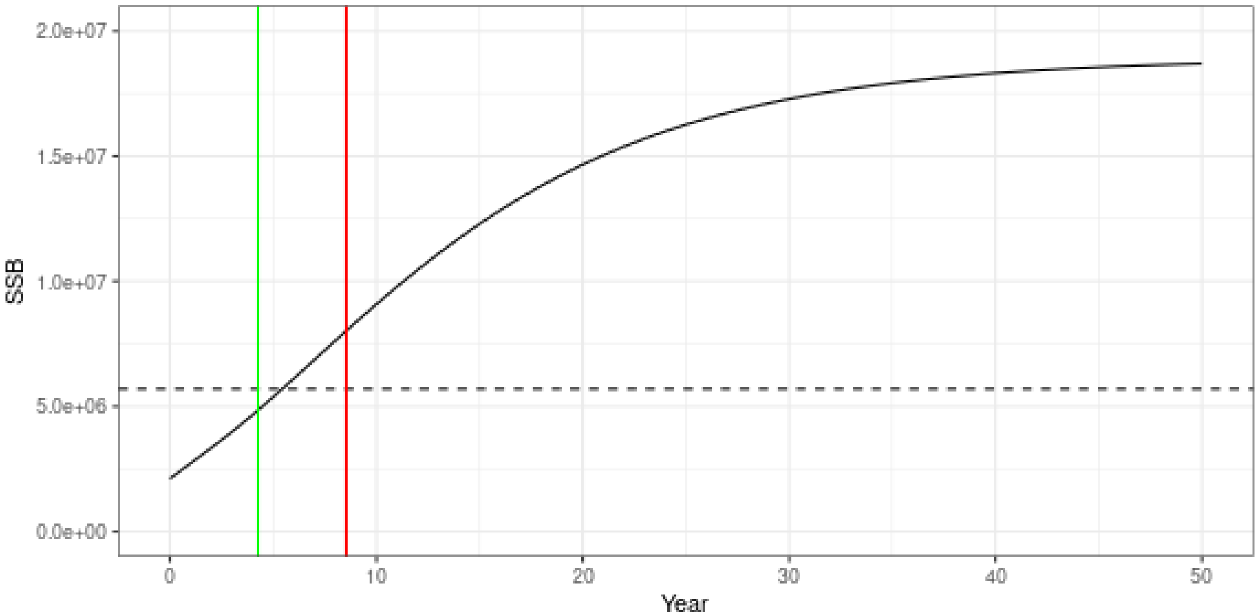


Figure 2 Recovery from depletion, vertical lines are half (green) and one (red) generation time

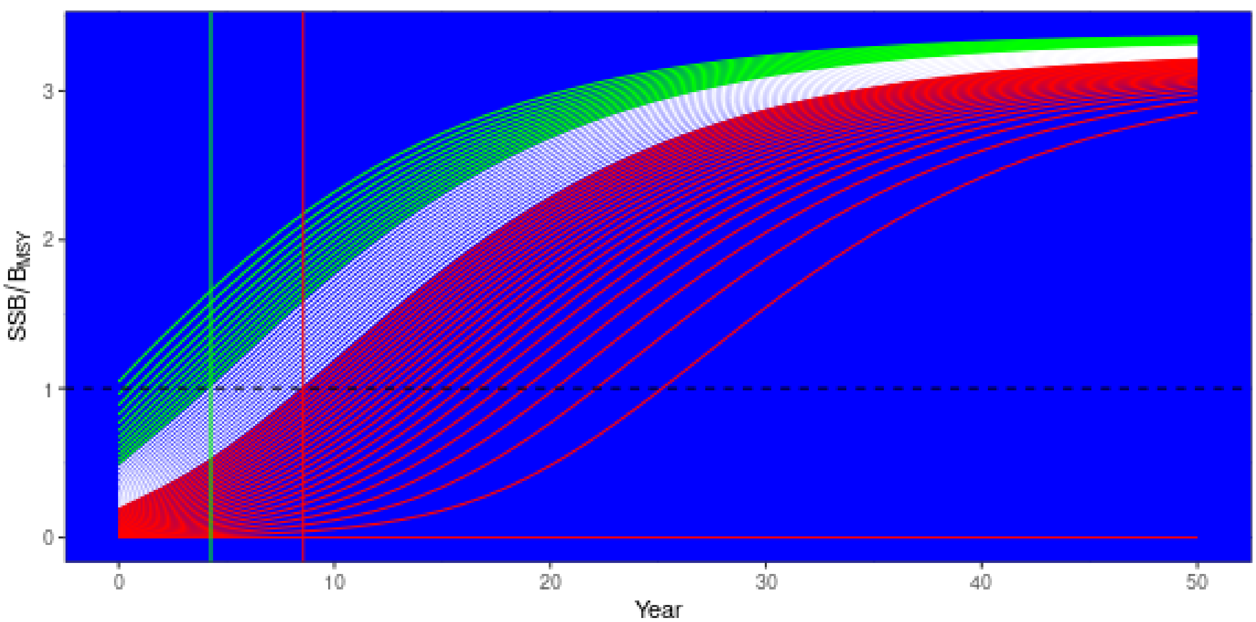


Figure 3 Recovery from different depletion levels, red horizontal line is B_{MSY}, vertical lines are half (green) and one (red) generation time

If the management objective is in the case of stock depletion is to close the fishery to recover a stock in a generation time then the biomass limit that should be avoided with high probability is 25% of B_{MSY}. If fishing is still to be allowed under recovery then the biomass limit will have to be higher.

As a performance statistic you could then say there must be a 95% prob that you can rebuild within a given fraction of a generation time with F=0.

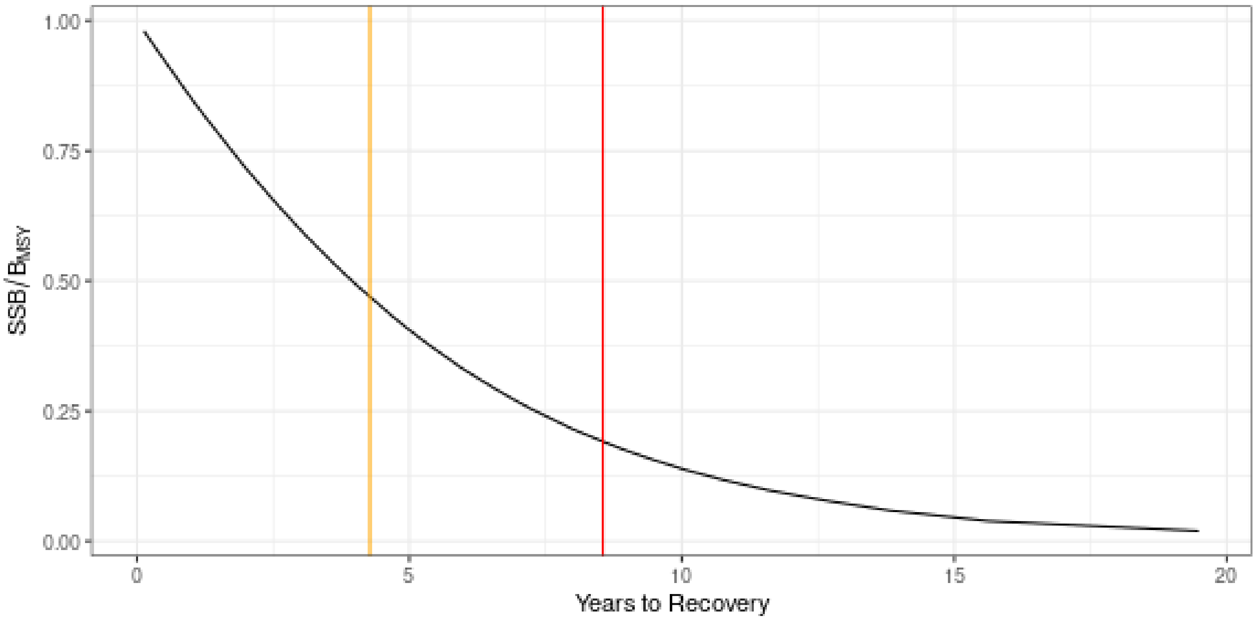
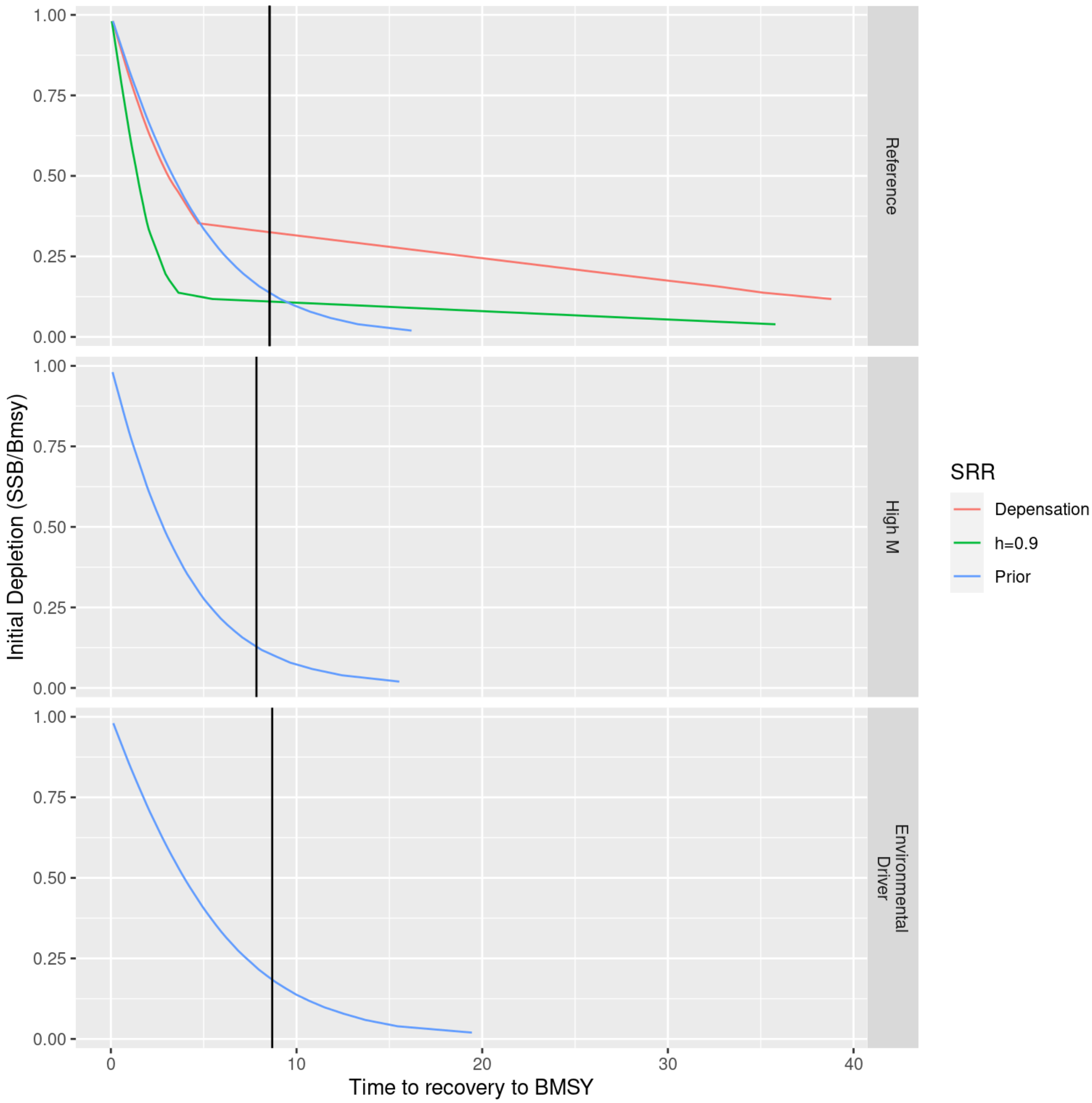


Figure 4 Time to recover from different depletion levels, vertical lines are half (white) and one (red) generation time

Impact of M and the SRR



References

Hilton-Taylor, Craig, and David Brackett. 2000. "2000 IUCN Red List of Threatened Species."

Mace, Georgina M, Nigel J Collar, Kevin J Gaston, CRAIG Hilton-Taylor, H Resit Akçakaya, NIGEL Leader-Williams, Eleanor Jane Milner-Gulland, and Simon N Stuart. 2008. "Quantification of Extinction Risk: IUCN's System for Classifying Threatened Species." *Conservation Biology* 22 (6): 1424–42.